SUMMATIVE ASSESSMENT – 1 (SA-1) MODEL PAPER – SET 3

CLASS X · MATHEMATICS · AP State Syllabus 2026-27 · SSC Blueprint 15E & 16E Pattern

BLUEPRINT – SA-1 MODEL PAPER – SET 3

SUBJECT: MATHEMATICS	CLASS: X	TIME: 3 Hrs 15 Min	MAX. MARKS: 100

Syllabus: Chapters 1 – 11 of Class X Mathematics (AP SCERT, 2026-27) – Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations, Arithmetic Progressions, Triangles, Coordinate Geometry, Statistics, Introduction to Trigonometry, Applications of Trigonometry, Probability. Pattern: SSC Public Examinations 2026-27 Blueprint & Model Paper (15E & 16E).

1. Weightage to Academic Standards (Ability)

ABILITY	AWARENESS (60%)	SENSITIVITY (20%)	CREATIVITY (20%)	TOTAL
MARKS	60	20	20	100

2. Weightage to Form of Questions

FORM OF QUESTION	Essay / Long Ans.	Short Answer	Very Short Ans.	Objective	TOTAL
No. of questions	5	8	8	12	33
Marks allotted	8	4	2	1	100
Estimated time	70 min	60 min	30 min	20 min	180 + 15 min

3. Ability-wise Distribution of Questions

ABILITY	1 Mark	2 Marks	4 Marks	8 Marks	Total Marks
Awareness	4	4	6	3	60
Sensitivity	4	2	1	1	20
Creativity	4	2	1	1	20
Total Questions	12	8	8	5	100

4. Chapter-wise Weightage

S.No	UNIT / CHAPTER	MARKS
1	Real Numbers	1+8 = 9
2	Polynomials	1+2+4 = 7
3	Pair of Linear Equations	1+2+8 = 11
4	Quadratic Equations	1+2+4+(8) = 7 + (8)
5	Arithmetic Progressions	1+4+(8) = 5 + (8)
6	Triangles	1+1+2+4+(8) = 8 + (8)
7	Introduction to Trigonometry	1+2+4+(8) = 7 + (8)
8	Applications of Trigonometry	1+2+(8) = 3 + (8)
9	Coordinate Geometry	1+2+4+8 = 15
10	Statistics	1+2+4+8 = 15
11	Probability	1+4+8 = 13
	TOTAL	**100 + (40)**

Note: Marks within ( ) are the internal-choice (OR) questions of Section IV. The marks of the chapters that are not part of SA-1 in the SSC blueprint (Circles, Areas Related to Circles, Surface Areas & Volumes) are redistributed among Chapters 1 – 11; the ability-wise (60 : 20 : 20) and form-wise weightage are exactly as in the SSC blueprint.

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5. Question-wise Chapter Distribution and Domain (Questions 1 to 33)

Q.No	Chapter / Topic	Domain	Marks
1	Real Numbers	Awareness	1
2	Polynomials	Creativity	1
3	Pair of Linear Equations	Creativity	1
4	Quadratic Equations	Creativity	1
5	Arithmetic Progressions	Sensitivity	1
6	Triangles	Awareness	1
7	Introduction to Trigonometry	Sensitivity	1
8	Applications of Trigonometry	Sensitivity	1
9	Coordinate Geometry	Creativity	1
10	Statistics	Awareness	1
11	Probability	Awareness	1
12	Triangles	Sensitivity	1
13	Polynomials	Creativity	2
14	Quadratic Equations	Sensitivity	2
15	Triangles	Awareness	2
16	Coordinate Geometry	Sensitivity	2
17	Introduction to Trigonometry	Awareness	2
18	Statistics	Awareness	2
19	Pair of Linear Equations	Awareness	2
20	Applications of Trigonometry	Creativity	2
21	Polynomials	Awareness	4
22	Quadratic Equations	Awareness	4
23	Arithmetic Progressions	Awareness	4
24	Introduction to Trigonometry	Awareness	4
25	Triangles	Sensitivity	4
26	Coordinate Geometry	Awareness	4
27	Statistics	Awareness	4
28	Probability	Creativity	4
29a / 29b	(a) Real Numbers (b) Arithmetic Progressions	Creativity	8
30a / 30b	(a) Statistics (b) Triangles	Awareness	8
31a / 31b	(a) Coordinate Geometry (b) Introduction to Trigonometry	Sensitivity	8
32a / 32b	(a) Probability (b) Quadratic Equations	Awareness	8
33a / 33b	(a) Pair of Linear Equations (b) Applications of Trigonometry	Awareness	8

ABILITIES AND THEIR OBJECTIVES AS PER PARAKH NORMS

I) AWARENESS: Knowledge / understanding of the construct / understanding of the process (resembles old objectives Knowledge, Understanding and Application).

II) SENSITIVITY: Conflict resolution / open-mindedness / collaboration (resembles old objective Analysis).

III) CREATIVITY: Generate innovative ideas / flexibility / fluency / exploration / combining ideas and concepts (resembles old objectives Evaluation and Creativity).

NOTE: Overall, the chapter-wise and ability-wise weightage is followed. There is no rigid rule that a question with a particular number must come from a particular chapter.

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SUMMATIVE ASSESSMENT – 1 : MODEL QUESTION PAPER – SET 3

MATHEMATICS – CLASS X (English Medium, AP State Syllabus 2026-27)

Time : 3 Hrs 15 Min (15 min reading + 3 hrs writing) **Max. Marks : 100**

General Instructions: *(i) Answer all the questions. (ii) Write the answers neatly in the answer booklet. (iii) The paper has four Sections (I, II, III, IV). (iv) Use of calculators is not permitted. (v) Draw neat, labelled figures wherever necessary.*

SECTION – I 12 × 1 = 12 Marks

Note : *i) Answer all the questions. ii) Each question carries 1 mark. iii) Choose the correct option and write its letter (A/B/C/D) in the answer booklet.*

1. The product of the HCF and LCM of 12 and 18 is *[Awareness · Real Numbers]*

(A) 6 (B) 36 (C) 216 (D) 72

2. If 2 and −3 are the zeroes of the polynomial $x^2 + (a + 1)x + b$, then (a, b) = *[Creativity · Polynomials]*

(A) (−2, −6) (B) (2, −6) (C) (0, −6) (D) (0, 6)

3. The pair of linear equations $kx - y = 2$ and $6x - 2y = 3$ has a unique solution if *[Creativity · Pair of Linear Equations]*

(A) $k = 3$ (B) $k \neq 3$ (C) $k \neq 0$ (D) $k = 0$

4. The quadratic equation $2x^2 - \sqrt{5}\,x + 1 = 0$ has *[Creativity · Quadratic Equations]*

(A) two distinct real roots (B) two equal real roots (C) no real roots (D) more than two real roots

5. The first row of a hall has 20 seats and every next row has 2 more seats than the previous row. The number of seats in the 12th row is *[Sensitivity · Arithmetic Progressions]*

(A) 40 (B) 42 (C) 44 (D) 46

6. If $\Delta ABC \sim \Delta DEF$, $\angle A = 47°$ and $\angle E = 83°$, then $\angle C$ = *[Awareness · Triangles]*

(A) 50° (B) 60° (C) 40° (D) 70°

7. $(1 + \tan^2\theta)(1 - \sin\theta)(1 + \sin\theta)$ = *[Sensitivity · Introduction to Trigonometry]*

(A) 0 (B) 1 (C) 2 (D) $\sin^2\theta$

8. A kite is flying with its 50 m long string inclined at 30° to the ground (no slack). The height of the kite is *[Sensitivity · Applications of Trigonometry]*

(A) 25 m (B) $25\sqrt{3}$ m (C) 50 m (D) $50\sqrt{3}$ m

9. If the centroid of the triangle with vertices (3, −5), (−7, 4) and (10, −k) is (2, −1), then k = *[Creativity · Coordinate Geometry]*

(A) 2 (B) −2 (C) 3 (D) 4

10. In the median formula $l + ((N/2 - cf)/f) \times h$, 'cf' stands for *[Awareness · Statistics]*

(A) frequency of the median class (B) cumulative frequency of the class preceding the median class

(C) class size (D) cumulative frequency of the median class

11. A card is selected at random from cards numbered 1 to 10. The probability that the number on it is divisible by 3 is *[Awareness · Probability]*

(A) 1/10 (B) 3/10 (C) 1/5 (D) 2/5

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12. A vertical stick 6 m long casts a shadow 4 m long. At the same time a tower casts a shadow 28 m long. The height of the tower is *[Sensitivity · Triangles]*

(A) 36 m (B) 42 m (C) 48 m (D) 24 m

SECTION – II 8 × 2 = 16 Marks

Note : *i) Answer all the questions. ii) Each question carries 2 marks.*

13. If α and β are the zeroes of the polynomial $x^2 - 5x + k$ such that $\alpha - \beta = 1$, find the value of k. *[Creativity · Polynomials]*

14. For what value of k does the equation $kx(x - 2) + 6 = 0$ have two equal roots? *[Sensitivity · Quadratic Equations]*

15. State the Pythagoras theorem. *[Awareness · Triangles]*

16. Find the point on the x-axis which is equidistant from the points (2, −5) and (−2, 9). *[Sensitivity · Coordinate Geometry]*

17. If tan A = 5/12, find the values of sin A and sec A. *[Awareness · Introduction to Trigonometry]*

18. Write the empirical relation connecting mean, median and mode. If the mean is 36 and the median is 34, find the mode. *[Awareness · Statistics]*

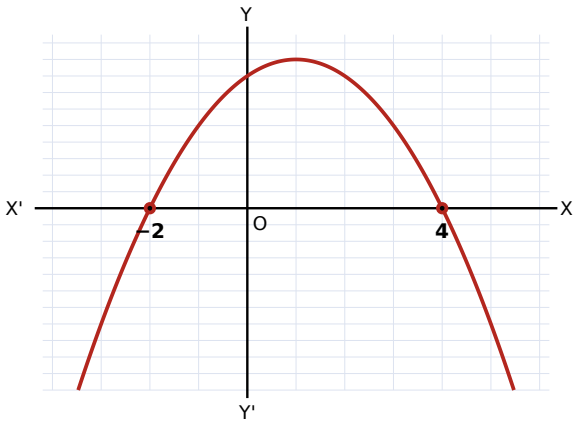
19. Solve by the substitution method: x + 2y = 7 and 3x − y = 7. *[Awareness · Pair of Linear Equations]*

20. The angle of depression of a boat from the top of a 30 m high cliff is 30°. Draw a rough figure and find the distance of the boat from the foot of the cliff. *[Creativity · Applications of Trigonometry]*

SECTION – III 8 × 4 = 32 Marks

Note : *i) Answer all the questions. ii) Each question carries 4 marks.*

21. Observe the graph of a polynomial given below and answer the questions. *[Awareness · Polynomials]*



i) How many zeroes does the polynomial have?
ii) Write the zeroes of the polynomial.
iii) Does the parabola open upward or downward? What does it tell about the sign of 'a' in $ax^2 + bx + c$?
iv) Find the sum and the product of the zeroes.

22. Solve the quadratic equation $2x^2 - 5x + 3 = 0$ by the method of completing the square. *[Awareness · Quadratic Equations]*

23. The first term of an AP is 17, the last term is 350 and the common difference is 9. How many terms are there and what is their sum? *[Awareness · Arithmetic Progressions]*

24. Prove that: (tan A + sin A)/(tan A − sin A) = (sec A + 1)/(sec A − 1). *[Awareness · Introduction to Trigonometry]*

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25. Two poles of heights 6 m and 11 m stand vertically on a plane ground. If the distance between their feet is 12 m, find the distance between their tops. *[Sensitivity · Triangles]*

26. Find the coordinates of the point which divides the line segment joining A(−1, 7) and B(4, −3) in the ratio 2 : 3. *[Awareness · Coordinate Geometry]*

27. The marks obtained by 40 students in a test are given below. Find the median marks. *[Awareness · Statistics]*

Marks	**0–20**	**20–40**	**40–60**	**60–80**	**80–100**
No. of students	5	8	14	9	4

28. Three coins are tossed simultaneously. Write any four different events based on this experiment and find the probability of each. *[Creativity · Probability]*

SECTION – IV 5 × 8 = 40 Marks

Note : *i) Answer all the questions. ii) Each question carries 8 marks. iii) There is an internal choice for each question.*

29.

a) Use Euclid's division algorithm to find the HCF of 867 and 255 and express it in the form 867m + 255n. *[Creativity · Real Numbers]*

OR

b) A sum of Rs. 1050 is to be used to give six cash prizes to students of a school for their overall academic performance. If each prize is Rs. 30 less than the preceding prize, find the value of each prize. *[Creativity · Arithmetic Progressions]*

30.

a) The following distribution gives the marks of 54 students. Find the mode and the median of the data. *[Awareness · Statistics]*

Marks	**0–10**	**10–20**	**20–30**	**30–40**	**40–50**	**50–60**
No. of students	4	7	12	20	8	3

OR

b) State and prove the theorem on the ratio of the areas of two similar triangles. *[Awareness · Triangles]*

31.

a) The vertices of a triangle are A(0, −1), B(2, 1) and C(0, 3). Find the area of the triangle formed by joining the mid-points of its sides. Also find the ratio of this area to the area of ΔABC and state what you observe. *[Sensitivity · Coordinate Geometry]*

OR

b) If sec θ + tan θ = p, prove that $(p^2 - 1)/(p^2 + 1) = \sin\theta$. *[Sensitivity · Introduction to Trigonometry]*

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32.

a) A bag contains 4 red, 5 blue, 6 green, 3 white and 2 yellow balls. One ball is drawn at random. Find the probability that it is: *[Awareness · Probability]*

i) green

ii) not blue

iii) red or yellow

iv) neither red nor white

OR

b) The length of a rectangular garden is 3 m more than its breadth. Its area is 6 m² more than the area of a triangular park of base 12 m and height 8 m. Find the length and breadth of the garden. *[Awareness · Quadratic Equations]*

33.

a) Draw the graph of the pair of linear equations $x - y = 1$ and $2x + 3y = 12$ and find the solution from the graph. *[Awareness · Pair of Linear Equations]*

OR

b) From a point on the ground, the angles of elevation of the bottom and the top of a transmission tower fixed at the top of a 20 m high building are 45° and 60° respectively. Find the height of the tower. *[Awareness · Applications of Trigonometry]*

— ALL THE BEST —

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